

Von Willebrand Disease Testing

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Abbreviations	
Activity:Ag	Ratio of VWF activity to VWF:Ag
FVIII	Factor VIII
n/a	Not available
RIPA	Ristocetin-induced platelet aggregation
VWD	Von Willebrand disease
VWF	Von Willebrand factor
VWF:Ag	Von Willebrand factor antigen

INDICATIONS FOR TESTING
Excessive mucosal bleeding (e.g., epistaxis, bleeding of gums, menorrhagia, postsurgical bleeding)

**PERFORM
Initial testing:**

- FVIII activity testing
- VWF activity testing
- VWF:Ag testing

CALCULATE

- Activity:Ag ratio

AND CONSIDER:

- An assay or panel that reflexes to multimeric analysis

Results compatible
with VWD

Results **not** compatible
with VWD

**PERFORM AS INDICATED
Additional testing to confirm subtype:**

- Multimeric analysis
- VWF gene sequencing
- RIPA test

Referral or repeat
testing in 1-3 mos
if high suspicion
for VWD

Type 1

- VWF:Ag: <30 IU/dL
- VWF activity: <30 IU/dL
- FVIII: low or normal
- Activity:Ag ratio: >0.5-0.7
- Multimer pattern: normal but reduced intensity
- RIPA: usually normal
- Type 1 variant present
- Consider desmopressin trial

Type 1C

- VWF:Ag: <30 IU/dL
- VWF activity: <30 IU/dL
- FVIII: low or normal
- Activity:Ag ratio: >0.5-0.7
- Multimer pattern: normal but reduced intensity
- RIPA: usually normal
- Type 1 variant present
- Desmopressin trial compatible with increased VWF clearance

Type 2A

- VWF:Ag: <30-200 (commonly <50) IU/dL
- VWF activity: <30 IU/dL
- FVIII: low or normal
- Activity:Ag ratio: <0.5-0.7
- Multimer pattern: abnormal
- RIPA: often reduced with high ristocetin concentrations; no enhanced aggregation with low ristocetin concentrations
- Type 2A variant present

Type 2B and Platelet Type

- VWF:Ag: <30-200 (commonly <50) IU/dL
- VWF activity: <30 IU/dL
- FVIII: low or normal
- Activity:Ag ratio: typically <0.5-0.7
- Multimer pattern: abnormal
- RIPA: abnormal; enhanced aggregation with low ristocetin concentrations
- Type 2B variant present

Type 2M

- VWF:Ag: <30-200 (commonly <50) IU/dL
- VWF activity: <30 IU/dL
- FVIII: low or normal
- Activity:Ag ratio: <0.5-0.7
- Multimer pattern: normal pattern, may have reduced intensity
- RIPA: reduced aggregation with high ristocetin concentrations
- Type 2M variant present

Type 2N

- VWF:Ag: 30-200 IU/dL
- VWF activity: 30-200 IU/dL
- FVIII: mildly or markedly low
- Activity:Ag ratio: >0.5-0.7
- Multimer pattern: normal pattern, may have reduced intensity
- RIPA: normal
- Type 2N variants present

Type 3

- VWF:Ag: absent
- VWF activity: absent
- FVIII: severely low (<10 IU/dL)
- Activity:Ag ratio: n/a
- Multimer pattern: no VWF present
- RIPA: absent aggregation
- Type 3 variants present

References

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- Platten S, Baker P, Bowyer A, et al. [Guideline for laboratory diagnosis and monitoring of von Willebrand disease: a joint guideline from the United Kingdom Haemophilia Centre Doctors' Organisation and the British Society for Haematology](#). *Br J Haematol*. 2024;204(5):1714-1731.